

LAB

OpenLANE Directory structure in detail

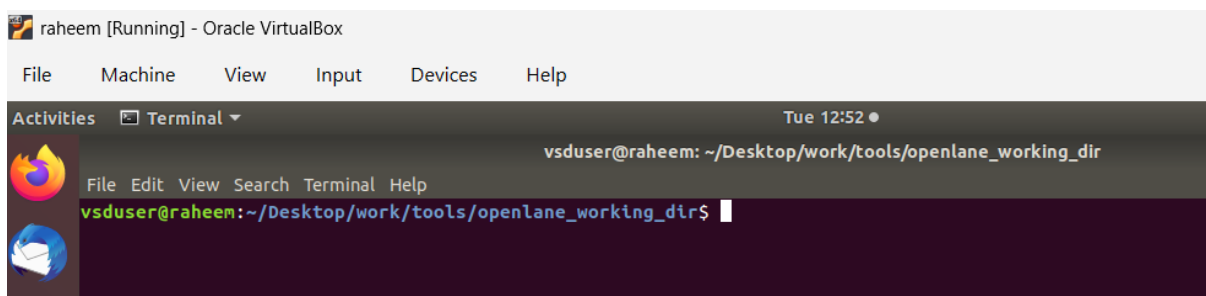
Lab Introduction

This workshop guides you through the setup and exploration of the **OpenLane flow**, an open-source toolchain for digital hardware design. The aim is to automate the process from RTL description all the way to GDSII layout using open-source tools and the SkyWater 130nm PDK. Key steps include navigating directories, exploring PDK contents, and invoking the OpenLane workflow.

Lab Steps

1. Start in Home Directory

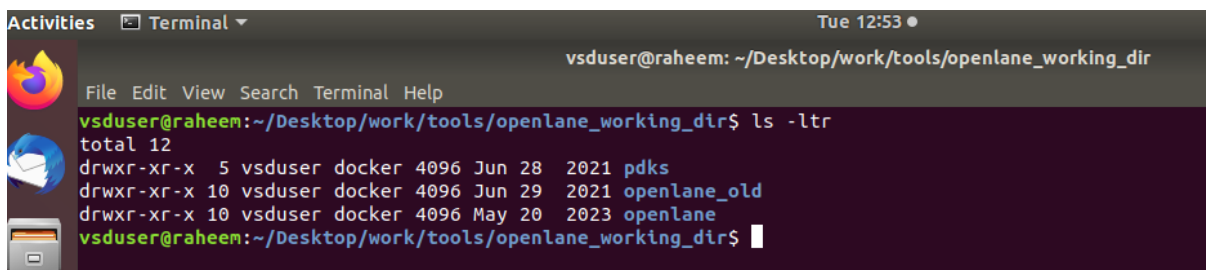
You will begin in your **Linux terminal** within your home directory.



List Contents Chronologically

List all files and directories inside your current folder using `ls` with the `-ltr` switch (lists files in chronological order).

```
ls -ltr
```



Explore Directory Structure

Navigate into the **openlane_working_dir** directory to access the OpenLane flow and PDK folders:

```
cd openlane_working_dir
```

Examine the PDKs Folder

Change into the pdks directory:

```
cd pdks
```

```
File Edit View Search Terminal Help
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir$ ls -ltr
total 12
drwxr-xr-x  5 vsduser docker 4096 Jun 28  2021 pdks
drwxr-xr-x 10 vsduser docker 4096 Jun 29  2021 openlane_old
drwxr-xr-x 10 vsduser docker 4096 May 20  2023 openlane
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir$ cd pdks
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks$ ls
open_pdks sky130A skywater-pdk
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks$
```

Explore SkyWater 130nm PDK

Change to the sky130A directory (the SkyWater 130nm variant made compatible with OpenLane):

```
cd sky130A
```

```
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks$ cd sky130A/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ ls
libs.ref libs.tech SOURCES
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$
```

```
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ cd libs.ref/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref$ ls
sky130_fd_io sky130_fd_sc_hd sky130_fd_sc_hs sky130_fd_sc_lp sky130_fd_sc_ms sky130_osu_s
sky130_fd_pr sky130_fd_sc_hdll sky130_fd_sc_hvl sky130_fd_sc_ls sky130_ml_xx_hd sky130_sram
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref$ ls -ltr
total 48
drwxr-xr-x 10 vsduser docker 4096 Jun 28  2021 sky130_osu_sc_t18
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_ms
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_ls
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_hs
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_hdll
drwxr-xr-x  8 vsduser docker 4096 Jun 28  2021 sky130_fd_pr
drwxr-xr-x  9 vsduser docker 4096 Jun 28  2021 sky130_sram_macros
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_hvl
drwxr-xr-x 11 vsduser docker 4096 Jun 28  2021 sky130_fd_io
drwxr-xr-x  4 vsduser docker 4096 Jun 28  2021 sky130_ml_xx_hd
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_lp
drwxr-xr-x 12 vsduser docker 4096 Jun 28  2021 sky130_fd_sc_hd
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref$
```

You should see libs.ref and libs.tech directories.

- **libs.ref** contains process-specific files (timing, liberty, cell info).

- **libs.tech** contains tool-specific files (layout engines, netgen, Magic overlays).

- **Inspect libs.ref**

- Go into libs.ref directory:
- `cd libs.ref`

```
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A$ cd libs.tech/  
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$ ls -ltr  
total 36  
drwxr-xr-x  9 vsduser docker 4096 Jun 28  2021 xschem  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 xcircuit  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 qflow  
drwxr-xr-x 10 vsduser docker 4096 Jun 28  2021 openlane  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 ngspice  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 netgen  
drwxr-xr-x  4 vsduser docker 4096 Jun 28  2021 magic  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 klayout  
drwxr-xr-x  2 vsduser docker 4096 Jun 28  2021 irsim  
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech$
```

To start the OpenLane tool, you will proceed from the openlane working directory.

Design Preparation Step

This lab guides you through performing an **interactive OpenLane flow** on a sample RISC-V core design. Key steps include launching the interactive session, inspecting and setting up the design folder, configuring design parameters, preparing the file system, and merging library files required for synthesis.

Lab Steps

1. Start OpenLane Interactive Session

Begin by running OpenLane in interactive mode. This allows you to execute each step of the flow manually, observe changes, and compare results.

docker

./flow.tcl -interactive

```

vsduser@raheem:~/Desktop/work/tools/openlane_working_dir$ docker
bash-4.2$ exit
exit
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir$ cd openlane
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane$ docker
bash-4.2$ ./flow.tcl -interactive
[INFO]:

  / \ | | \ / _] | | \ | | / \ | | \ / _]
  |  | | | ) [ _] | | | | | | | | | | | |
  |  | | | / [ _] | | | | | | | | | | | |
  |  | | | [ _] | | | | | | | | | | | |
  \_/ | | | | | | | | | | | | | | | |

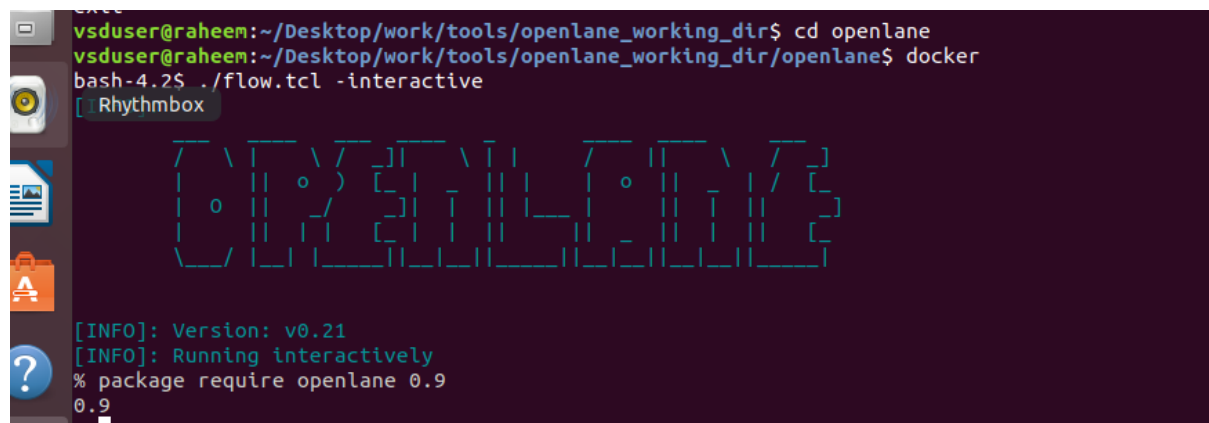
[INFO]: Version: v0.21
[INFO]: Running interactively
%

```

Load Required OpenLane Packages

Next, load the necessary OpenLane packages for the flow to run.

package require openlane 0.9



```

vsduser@raheem:~/Desktop/work/tools/openlane_working_dir$ cd openlane
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane$ docker
bash-4.2$ ./flow.tcl -interactive
[INFO]: Version: v0.21
[INFO]: Running interactively
% package require openlane 0.9
0.9
%

```

Explore Available Designs

OpenLane comes with a **designs** folder containing many example projects. Locate and explore the available designs.

cd designs

ls

```
% package require openlane 0.9
0.9
% ls
AUTHORS.md      Makefile      conf.py      designs      flow.tcl      run_d
CONTRIBUTING.md README.md     configuration docker_build regression_results scrip
LICENSE         clean_runs.tcl default.cvcrc docs          report_generation_wrapper.py
% cd designs
% ls
151      aes128      chacha      inverter      ocs_blitter
APU      aes192      cic_decimator jpeg_encoder  picorv32a
BM64     aes256      des         ldpc_decoder_802_3an point_add
PPU      aes_cipher des3         ldpcenc       point_scalar_mul
README.md aes_core   digital_pll_sky130_fd_sc_hd manual_macro_placement_test s44
aes      blabla     genericfir   md5           salsa20
% cd picorv32a
```

You will see three primary files:

- src/ — Contains the Verilog source file and SDC information.
- config.tcl — The PDK-specific configuration for this design.
- config.json — Optional; can bypass some configuration defaults.

The src directory typically contains the Verilog RTL file for the design along with its SDC (Synopsys Design Constraints).

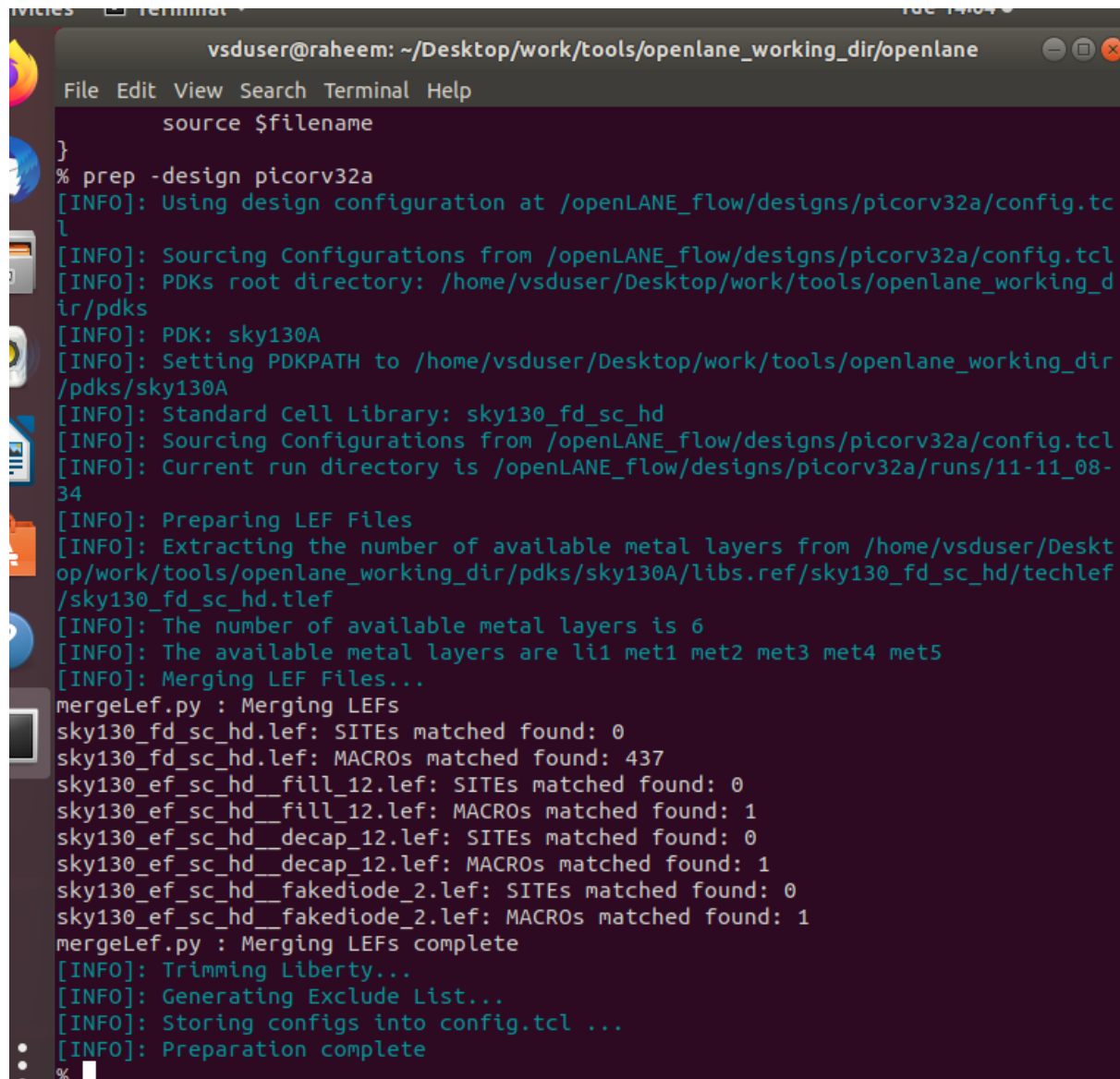
```
aes      blabla     genericfir   md5           salsa20
% cd picorv32a
couldn't change working directory to "picorv32a": no such file or directory
% cd picorv32a
% ls
config.tcl      sky130A_sky130_fd_sc_hdll_config.tcl  sky130A_sky130_fd_sc_ls
sky130A_sky130_fd_sc_hd_config.tcl  sky130A_sky130_fd_sc_hs_config.tcl  sky130A_sky130_fd_sc_ms
```


Prepare File System (Design Setup)

Return to the OpenLane prompt and prepare the file system for the flow. This step creates necessary directories and data structures for your design:

Note: Run the below command in openlane directory

```
prep -design picorv32a
```



```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane
File Edit View Search Terminal Help
source $filename
}
% prep -design picorv32a
[INFO]: Using design configuration at /openLANE_flow/designs/picorv32a/config.tcl
[INFO]: Sourcing Configurations from /openLANE_flow/designs/picorv32a/config.tcl
[INFO]: PDKs root directory: /home/vsduser/Desktop/work/tools/openlane_working_dir/pdks
[INFO]: PDK: sky130A
[INFO]: Setting PDKPATH to /home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Sourcing Configurations from /openLANE_flow/designs/picorv32a/config.tcl
[INFO]: Current run directory is /openLANE_flow/designs/picorv32a/runs/11-11_08-34
[INFO]: Preparing LEF Files
[INFO]: Extracting the number of available metal layers from /home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/techlef/sky130_fd_sc_hd.tlef
[INFO]: The number of available metal layers is 6
[INFO]: The available metal layers are li1 met1 met2 met3 met4 met5
[INFO]: Merging LEF Files...
mergeLef.py : Merging LEFs
sky130_fd_sc_hd.lef: SITES matched found: 0
sky130_fd_sc_hd.lef: MACROS matched found: 437
sky130_ef_sc_hd__fill_12.lef: SITES matched found: 0
sky130_ef_sc_hd__fill_12.lef: MACROS matched found: 1
sky130_ef_sc_hd__decap_12.lef: SITES matched found: 0
sky130_ef_sc_hd__decap_12.lef: MACROS matched found: 1
sky130_ef_sc_hd__fakediode_2.lef: SITES matched found: 0
sky130_ef_sc_hd__fakediode_2.lef: MACROS matched found: 1
mergeLef.py : Merging LEFs complete
[INFO]: Trimming Liberty...
[INFO]: Generating Exclude List...
[INFO]: Storing configs into config.tcl ...
[INFO]: Preparation complete
%
```

What happens in this step:

The preparation stage merges library (LEF) files.

It combines the cell-level and technology-level LEF files into one, so subsequent flow steps can easily access all necessary cell and layer definitions from a single place.

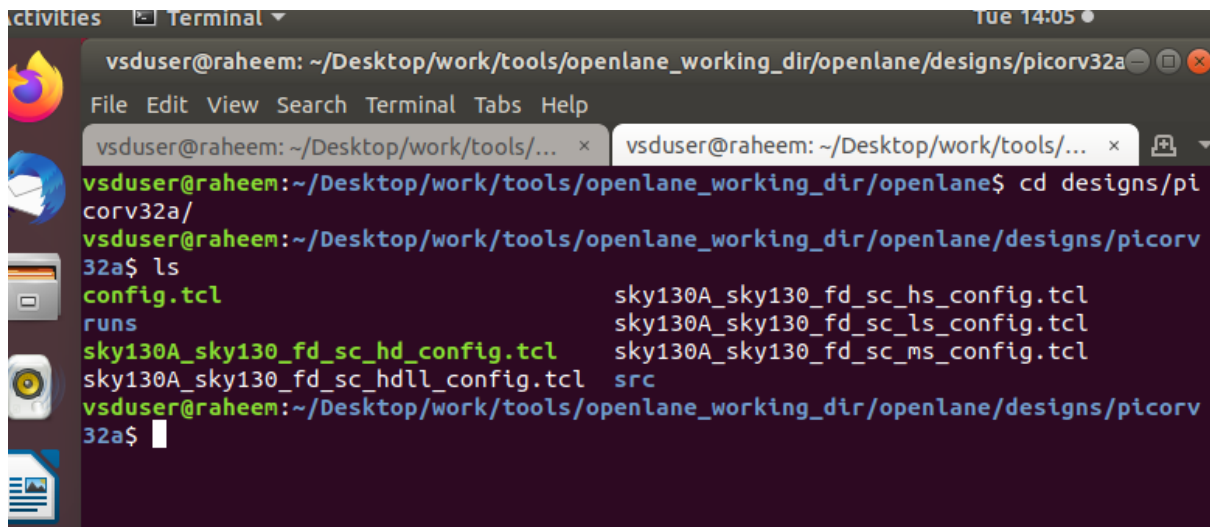
The directories and structure required by OpenLane for every flow stage are created.

You may see references to scripts such as mergeLef.py running in the output.

Verify Setup

Check if the preparation produced new directories and expected files inside your design directory.

ls

A terminal window titled 'Terminal' showing a user named 'vsduser' at a machine named 'raheem'. The user is in the directory '~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/'. They execute the command 'cd designs/picorv32a/' and then 'ls'. The output of 'ls' shows a directory named 'runs' and several .tcl files: 'sky130A_sky130_fd_sc_hd_config.tcl', 'sky130A_sky130_fd_sc_hdll_config.tcl', 'sky130A_sky130_fd_sc_hs_config.tcl', 'sky130A_sky130_fd_sc_ls_config.tcl', and 'sky130A_sky130_fd_sc_ms_config.tcl'. There is also a 'src' directory listed.

```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/
File Edit View Search Terminal Tabs Help
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane$ cd designs/picorv32a/
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a$ ls
runs                                sky130A_sky130_fd_sc_hs_config.tcl
sky130A_sky130_fd_sc_hd_config.tcl sky130A_sky130_fd_sc_ls_config.tcl
sky130A_sky130_fd_sc_hdll_config.tcl sky130A_sky130_fd_sc_ms_config.tcl
src
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a$
```

Navigate to the Runs Directory

After setup, your project will have a runs directory, with a subfolder named by date (e.g., 12th_August). Change into this directory to view the structure required by OpenLane.

```
cd runs/27-10_13-46
```

You can list the contents to see the standard folder layout:

```
tree
```

```

Processing triggers for libc-bin (2.35-0ubuntu3.1) ...
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ tree
.
├── cmds.log
├── config.tcl
├── logs
│   ├── 0-prep_runtime.txt
│   ├── cts
│   ├── cvc
│   ├── floorplan
│   ├── flow_summary.log
│   ├── klayout
│   ├── lvs
│   ├── magic
│   ├── placement
│   ├── routing
│   └── synthesis
├── OPENLANE_VERSION
├── PDK_SOURCES
├── reports
│   ├── cts
│   ├── cvc
│   ├── floorplan
│   ├── klayout
│   ├── lvs
│   ├── magic
│   ├── placement
│   ├── routing
│   └── synthesis

```

```

vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ tree
.
├── magic
├── placement
├── routing
├── synthesis
├── results
│   ├── cts
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── cvc
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── floorplan
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── klayout
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── lvs
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── magic
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── placement
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── routing
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   └── synthesis
│       └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── tmp
│   ├── cts
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── cvc
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── floorplan
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── klayout
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── lvs
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
│   ├── magic
│   │   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef

```

```
tmp
├── cts
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── cvc
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── floorplan
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── klayout
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── lvs
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── magic
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── merged.lef
├── merged_unpadded.lef
├── met_layers_list.txt
├── placement
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── routing
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── synthesis
│   └── merged_unpadded.lef -> ../../tmp/merged_unpadded.lef
├── tracks_copy.info
├── trimmed.lib
└── trimmed.lib.exclude.list

sduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$
```

```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane x vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/de..
DIRECTION INOUT ;
USE GROUND ;
PORT
    LAYER pwell ;
    RECT 0.145000 -0.085000 0.315000 0.085000 ;
END
END VNB
PIN VPB
DIRECTION INOUT ;
USE POWER ;
PORT
    LAYER nwell ;
    RECT -0.190000 1.305000 1.110000 2.910000 ;
END
END VPB
PIN VPWR
DIRECTION INOUT ;
SHAPE ABUTMENT ;
USE POWER ;
PORT
    LAYER met1 ;
    RECT 0.000000 2.480000 0.920000 2.960000 ;
END
END VPWR
OBS
    LAYER l11 ;
    RECT 0.000000 -0.085000 0.920000 0.085000 ;
    RECT 0.000000 2.635000 0.920000 2.805000 ;
    LAYER mcon ;
    RECT 0.145000 -0.085000 0.315000 0.085000 ;
    RECT 0.145000 2.635000 0.315000 2.805000 ;
    RECT 0.605000 -0.085000 0.775000 0.085000 ;
    RECT 0.605000 2.635000 0.775000 2.805000 ;
END
END sky130_ef_sc_hd__fakediode_2
END LIBRARYvsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$
```

xplore the tmp Folder

The tmp folder holds intermediate files generated during the flow. For example, after merging, you might find a file like merged.lef (created by merged.lef.py). Inspect its contents to see layer and cell-level information.

```
cat tmp/merged.lef
```

This file contains detailed layout information, such as wire-level data (at the top) and appended cell-level information (e.g., macro definitions).

Understand the Results, Reports, and logs Folders

- **results:** Contains output files for each stage (e.g., synthesis, placement, flow). Initially empty; populated as each stage completes.
- **reports:** Holds timing and other analysis reports if STC files are defined in your configuration. Also empty at the start.
- **logs:** Contains lock files for each step to prevent concurrent execution. The config.tcl file here (and sometimes in the design directory) is particularly important.

To view the current state of these folders:

```
ls -l results
```

```
ls -l reports
```

```
ls -l logs
```

```
END sky130_ef_sc_hd__taketode_2
END LIBRARYvsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ ls results/
cts cvc floorplan klayout lvs magic placement routing synthesis
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ ls
cmds.log config.tcl logs OPENLANE_VERSION PDK_SOURCES reports results tmp
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ ls reports/
cts cvc floorplan klayout lvs magic placement routing synthesis
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ ls logs
0-prep_runtime.txt cts cvc floorplan flow_summary.log klayout lvs magic placement routing synthesis
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$
```

Examine Conflict Article File

The config.tcl file shows which default parameters are being used in the current run, including PDK settings, track information, synthesis library, and timing libraries (min/max, transition). It is updated when configuration changes are made, providing a quick way to verify that your edits are reflected in the flow.

```
cat config.tcl
```

For example, if you change a parameter like CORE_UTILIZATION to 50 in your config and re-run the flow planning stage, this file will reflect the new value.

```
File Edit View Search Terminal Tabs Help
vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane x vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/de...
set ::env(openphysyn_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/placement/openphysyn"
set ::env(openphysyn_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/placement/openphysyn"
set ::env(opensta_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/synthesis/opensta"
set ::env(opensta_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/synthesis/opensta"
set ::env(opensta_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/synthesis/opensta"
set ::env(pdn_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/floorplan/pdn"
set ::env(pdn_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/floorplan/pdn"
set ::env(pdn_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/floorplan/pdn"
set ::env(replaceio_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/placement/replace"
set ::env(replaceio_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/placement/replace"
set ::env(replaceio_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/placement/replace"
set ::env(resizer_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/placement/resizer"
set ::env(resizer_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/placement/resizer"
set ::env(resizer_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/placement/resizer"
set ::env(tapcell_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/floorplan/tapcell"
set ::env(tapcell_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/floorplan/tapcell"
set ::env(tapcell_result_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/results/floorplan/picorv32a.floorplan"
set ::env(tapcell_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/floorplan/tapcell"
set ::env(tritonRoute_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/routing/tritonRoute"
set ::env(tritonRoute_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/routing/tritonRoute"
set ::env(tritonRoute_result_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/results/routing/picorv32a"
set ::env(tritonRoute_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/routing/tritonRoute"
set ::env(verilog2def_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/floorplan/verilog2def"
set ::env(verilog2def_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/floorplan/verilog2def"
set ::env(verilog2def_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/floorplan/verilog2def"
set ::env(yosys_log_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/logs/synthesis/yosys"
set ::env(yosys_report_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/reports/synthesis/yosys"
set ::env(yosys_result_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/results/synthesis/picorv32a.synthesis"
set ::env(yosys_tmp_file_tag) "/openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/synthesis/yosys"
set ::env(SYNTH_MAX_TRAN) "2.4730000000000003"
set ::env(CURRENT_DEF) "0"
set ::env(CURRENT_GUIDE) "0"
set ::env(CURRENT_INDEX) 0
set ::env(CURRENT_NETLIST) "0"
set ::env(PREV_NETLIST) "0"
```

```
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/runs/11-11_08-34$ cat cmds.log
Tue Nov 11 08:34:03 UTC 2025 - Executing "python3 /openLANE_flow/scripts/extract_metal_layers.py -t /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/techlef/sky130_fd_sc_hd.tlef -o /openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/met_layers_list.txt"

Tue Nov 11 08:34:04 UTC 2025 - Executing "/openLANE_flow/scripts/mergeLeF.py -i /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/techlef/sky130_fd_sc_hd.tlef {/home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lef/sky130_fd_sc_hd.tlef /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_ef_sc_hd/lef/sky130_ef_sc_hd.tlef /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lef/sky130_ef_sc_hd_decap.tlef /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lef/sky130_ef_sc_hd_fakediode_2.tlef} -o /openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/merged_unpadded.lef |& tee >&@stdout"

Tue Nov 11 08:34:04 UTC 2025 - Executing "/openLANE_flow/scripts/libtrim.pl /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd_tt_025C_1v80.lib /openLANE_flow/designs/picorv32a/runs/11-11_08-34/tmp/trimmed.lib"

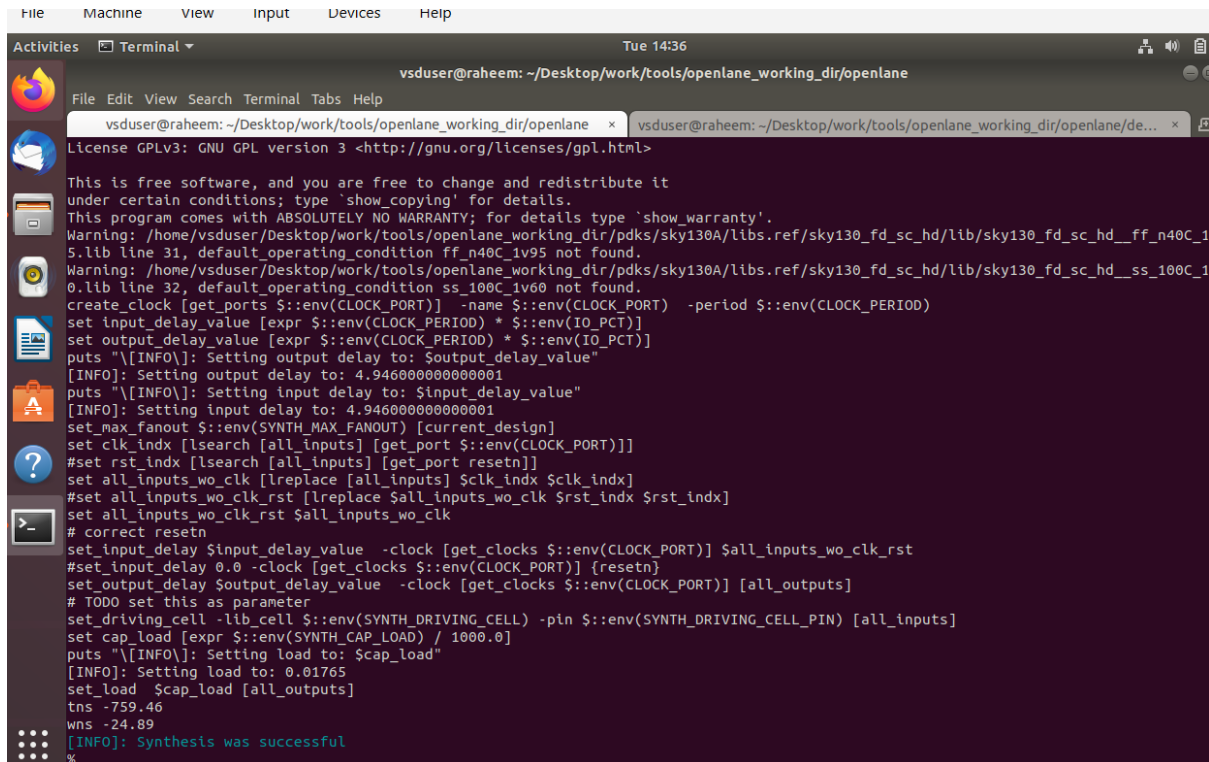
Tue Nov 11 08:34:04 UTC 2025 - Executing "echo {openlane v0.21} > /openLANE_flow/designs/picorv32a/runs/11-11_08-34/OPENLANE_VERSION"
```

Launch Synthesis

With the environment prepared, initiate the synthesis stage, which includes RTL synthesis and ABC optimization. This process typically takes 3–5 minutes.

run_synthesis

While synthesis runs, you can monitor progress in the terminal or by checking the results/synthesis directory as files are generated.

A screenshot of a terminal window on a Linux system. The terminal shows the output of a synthesis command. It starts with a license notice for GNU GPL version 3. Then, it displays several warnings about missing files in the library path. The main part of the output shows the synthesis process, including setting delays, fanouts, and loads. The final line indicates that the synthesis was successful.

```
File Machine View Input Devices Help
Activities Terminal
vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane
File Edit View Search Terminal Tabs Help
vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane x vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/de... x
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>

This is free software, and you are free to change and redistribute it
under certain conditions; type 'show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type 'show_warranty'.
Warning: /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd_ff_n40C_1
5.lib line 31, default_operating_condition ff_n40C_1v95 not found.
Warning: /home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd_ss_100C_1
0.lib line 32, default_operating_condition ss_100C_1v60 not found.
create_clock [get_ports $::env(CLOCK_PORT)] -name $::env(CLOCK_PORT) -period $::env(CLOCK_PERIOD)
set input_delay_value [expr $::env(CLOCK_PERIOD) * $::env(IO_PCT)]
set output_delay_value [expr $::env(CLOCK_PERIOD) * $::env(IO_PCT)]
puts "[INFO]: Setting output delay to: $output_delay_value"
[INFO]: Setting output delay to: 4.9460000000000001
puts "[INFO]: Setting input delay to: $input_delay_value"
[INFO]: Setting input delay to: 4.9460000000000001
set_max_fanout $::env(SYNTH_MAX_FANOUT) [current_design]
set clk_idx [lsearch [all_inputs] [get_port $::env(CLOCK_PORT)]]
#set rst_idx [lsearch [all_inputs] [get_port resetn]]
set all_inputs_wo_clk [lreplace [all_inputs] $clk_idx $clk_idx]
#set all_inputs_wo_clk_rst [lreplace $all_inputs_wo_clk $rst_idx $rst_idx]
set all_inputs_wo_clk_rst $all_inputs_wo_clk
# correct resetn
set input_delay $input_delay_value -clock [get_clocks $::env(CLOCK_PORT)] $all_inputs_wo_clk_rst
#set input_delay 0.0 -clock [get_clocks $::env(CLOCK_PORT)] {resetn}
set output_delay $output_delay_value -clock [get_clocks $::env(CLOCK_PORT)] [all_outputs]
# TODO set this as parameter
set_driving_cell -lib cell $::env(SYNTH_DRIVING_CELL) -pin $::env(SYNTH_DRIVING_CELL_PIN) [all_inputs]
set_cap_load [expr $::env(SYNTH_CAP_LOAD) / 1000.0]
puts "[INFO]: Setting load to: $cap_load"
[INFO]: Setting load to: 0.01765
set_load $cap_load [all_outputs]
tns -759.46
wns -24.89
[INFO]: Synthesis was successful
%
```

tips To Characterize Synthesis Results

This lab focuses on post-synthesis analysis in a digital hardware workflow, specifically aiming to compute the **flop ratio**, verify the synthesis output, and inspect related synthesis statistics and reports. The key steps are:

- Locate and interpret synthesis statistics to find D flip-flop counts.
- Calculate the flop ratio using cell counts from reports.
- Explore synthesis output directories for netlists and reports.
- Review how results are presented and interpreted in the context of the workflow.

Lab Steps

1. Find D Flip-Flop Count in Synthesis Statistics

- Open the synthesis statistics report provided by your synthesis tool.
- Look for the final D flip-flop (e.g., the cell type dfxtp_2) count and the total number of cells.

Example values:

- D flip-flop count: **1613**

- Total cell count: **14876**

```
vsduser@raheem: ~/Desktop/work/tools/... × vsduser@raheem: ~/Desktop/work/tools/... ×
28. Printing statistics.

=== picorv32a ===

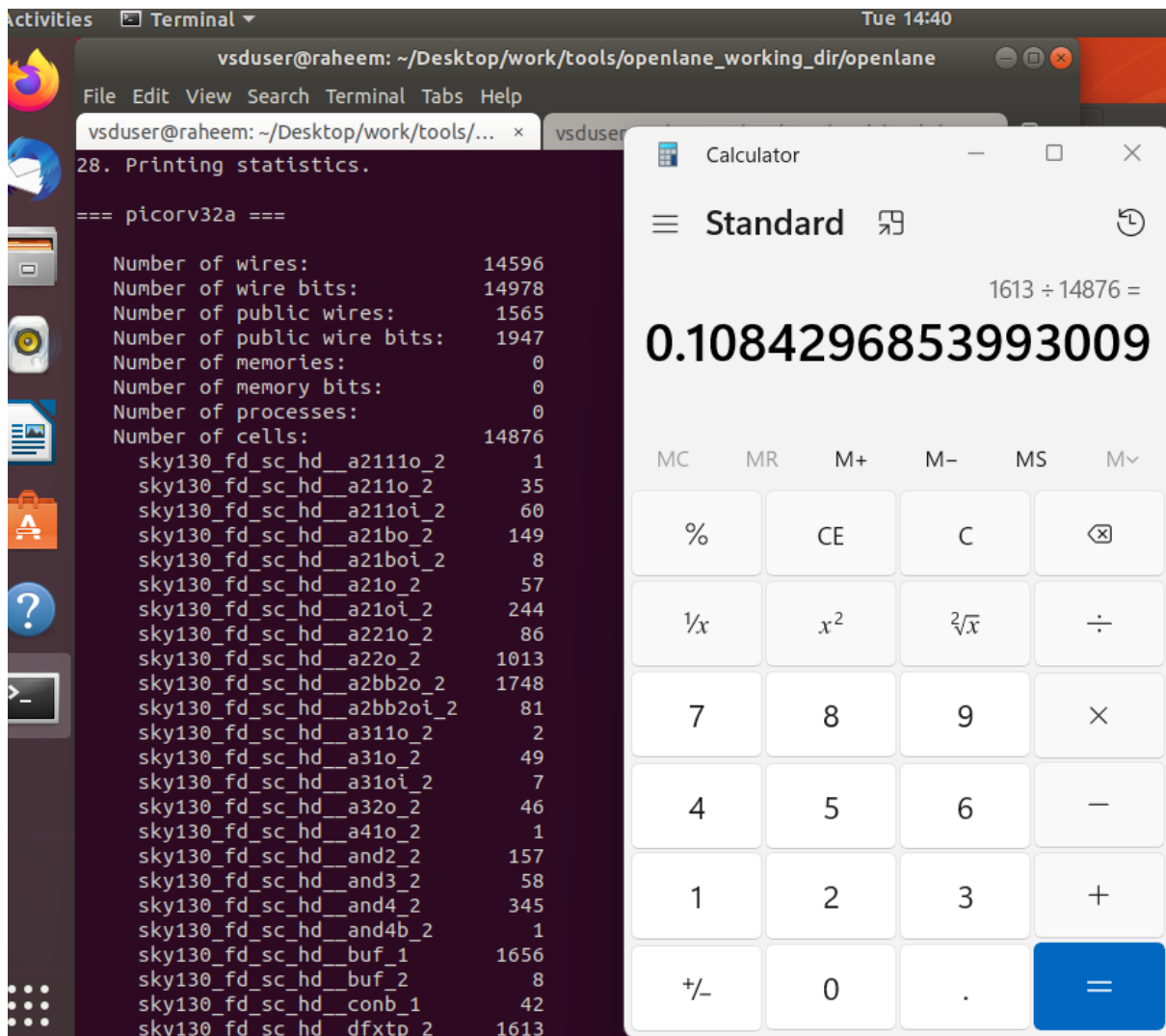
Number of wires:          14596
Number of wire bits:      14978
Number of public wires:   1565
Number of public wire bits: 1947
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          14876
sky130_fd_sc_hd__a2111o_2  1
sky130_fd_sc_hd__a211o_2   35
sky130_fd_sc_hd__a211oi_2  60
sky130_fd_sc_hd__a21bo_2   149
sky130_fd_sc_hd__a21boi_2   8
sky130_fd_sc_hd__a21o_2    57
sky130_fd_sc_hd__a21oi_2   244
sky130_fd_sc_hd__a221o_2   86
sky130_fd_sc_hd__a22o_2   1013
sky130_fd_sc_hd__a2bb2o_2  1748
sky130_fd_sc_hd__a2bb2oi_2  81
sky130_fd_sc_hd__a311o_2   2
sky130_fd_sc_hd__a31o_2    49
sky130_fd_sc_hd__a31oi_2   7
sky130_fd_sc_hd__a32o_2    46
sky130_fd_sc_hd__a41o_2    1
sky130_fd_sc_hd__and2_2    157
sky130_fd_sc_hd__and3_2    58
sky130_fd_sc_hd__and4_2    345
sky130_fd_sc_hd__and4b_2   1
sky130_fd_sc_hd__buf_1     1656
sky130_fd_sc_hd__buf_2     8
sky130_fd_sc_hd__conb_1    42
sky130_fd_sc_hd__dfxtp_2   1613
```

Calculate Flop Ratio

- Use a calculator or your preferred method to compute the flop ratio as:
- $\text{flop_ratio} = \text{D_flip_flop_count} / \text{Total_cell_count}$

For the given example values:

$$\text{flop_ratio} = 1613 / 14876$$



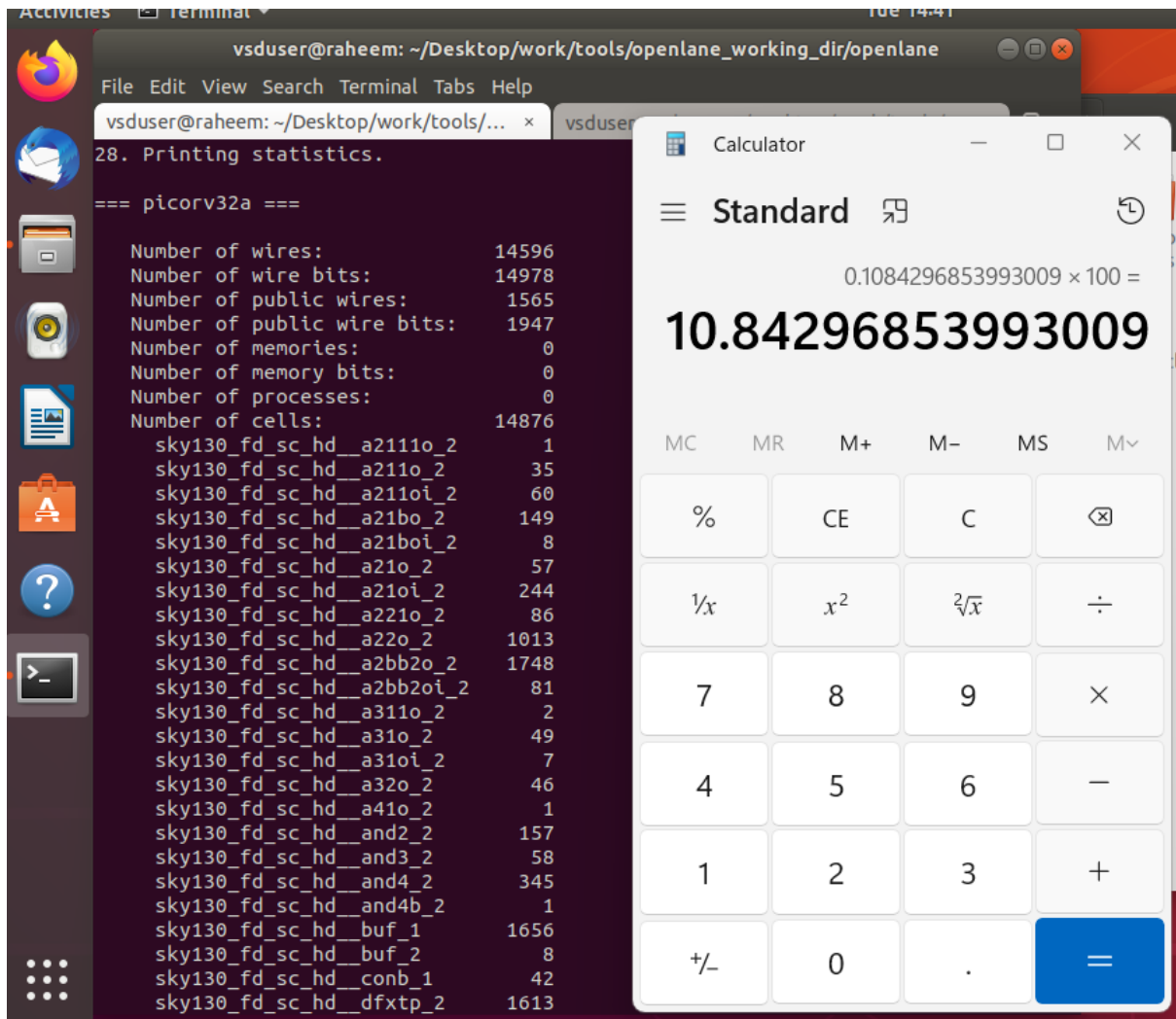
The output is:

0.108429

- To express as a percentage:
- $\text{flop_ratio_percent} = \text{flop_ratio} * 100$

So:

$\text{flop_ratio_percent} = 10.8429\%$



Locate Synthesis Results and Netlist

- Navigate to the results directory (typically named results or similar).
- Confirm that synthesis has completed and output files (e.g., netlist files) are present.
- cd results

ls

```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32...
File Edit View Search Terminal Help
32a/runs$ cd 11-11_08-54
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54$ ls
cmds.log      logs          PDK_SOURCES  results
config.tcl    OPENLANE_VERSION  reports      tmp
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54$ cd results/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54/results$ ls
cts  cvc  floorplan  klayout  lvs  magic  placement  routing  synthesis
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54/results$ ls -ltr
total 36
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 routing
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 placement
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 magic
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 lvs
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 klayout
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 floorplan
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 cvc
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:24 cts
drwxr-xr-x 2 vsduser vsduser 4096 Nov 11 14:25 synthesis
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54/results$
```

- The last file listed (by modification time) typically contains the most recent synthesis statistics.
- `ls -ltr`

`cat 1-yosys_4.stat.rpt`

```

vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32...
File Edit View Search Terminal Help

sky130_fd_sc_hd__o21bai_2      1
sky130_fd_sc_hd__o221a_2     204
sky130_fd_sc_hd__o221ai_2      7
sky130_fd_sc_hd__o22a_2     1312
sky130_fd_sc_hd__o22ai_2      59
sky130_fd_sc_hd__o2bb2a_2     119
sky130_fd_sc_hd__o2bb2ai_2     92
sky130_fd_sc_hd__o311a_2      8
sky130_fd_sc_hd__o31a_2      19
sky130_fd_sc_hd__o31ai_2      1
sky130_fd_sc_hd__o32a_2     109
sky130_fd_sc_hd__o41a_2      2
sky130_fd_sc_hd__or2_2     1088
sky130_fd_sc_hd__or2b_2      25
sky130_fd_sc_hd__or3_2      68
sky130_fd_sc_hd__or3b_2      5
sky130_fd_sc_hd__or4_2      93
sky130_fd_sc_hd__or4b_2      6
sky130_fd_sc_hd__or4bb_2      2

Chip area for module '\picorv32a': 147712.918400

vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
2a/runs/11-11_08-54/reports/synthesis$

```

Review Pre-Layout (OpenSTR) and Other Reports

- Optionally, check the open STR (Static Timing Report) or other pre-layout analysis reports in the relevant directory.

```

vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32...
File Edit View Search Terminal Help

      0.14    0.00    48.20 v _13751_/B (sky130_fd_sc_hd__or2_2)
      0.14    0.68    48.88 v _13751_/X (sky130_fd_sc_hd__or2_2)
2      0.01    10971_ (net)
      0.14    0.00    48.88 v _13754_/B2 (sky130_fd_sc_hd__o221a_2)
      0.07    0.44    49.33 v _13754_/X (sky130_fd_sc_hd__o221a_2)
1      0.00    03928_ (net)
      0.07    0.00    49.33 v _27762_/D (sky130_fd_sc_hd__dfxtp_2)
      49.33    data arrival time

      0.00    24.73    24.73 clock clk (rise edge)
      0.00    24.73    24.73 clock network delay (ideal)
      0.00    24.73    24.73 clock reconvergence pessimism
      24.73    ^ _27762_/CLK (sky130_fd_sc_hd__dfxtp_2)
      -0.29    24.44    library setup time
      24.44    data required time

-----
      24.44    data required time
     -49.33    data arrival time
-----
     -24.89    slack (VIOLATED)

vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_08-54/reports/synthesis$

```